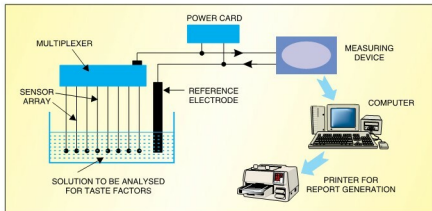


found to be the primary gateway in mammals for detection of spoiled and unripe food. These receptors are found in a subpopulation of taste sensor cells of tongue that do not function in sweet, bitter or umami. Taste researchers report that sour taste sensors can be effectively used to detect such spoiled food.

Sweet sensor is a small, cheap lab-on-a-chip sensor that quickly identifies sweetness—one of the five primary tastes.

Advantages of electronic tongues

1. Reduced measurement time
2. Less skill is required to operate the sensors
3. Required sample volume is small
4. Compact and robust in construction
5. High accuracy



Working principle of electronic taste sensor

6. Taste patterns obtained by electronic tongues give valuable and more information than the manual method

Applications

1. Analyse flavour ageing in beverages
2. Check the quality of bitterness

or spicy level of drinks or dissolved compounds

3. Check the quality of taste-masking efficiency of formulations (tablets, syrups, powders, capsules etc)
4. Analyse medicine stability in terms of taste
5. Provide benchmark for target products
6. Detect undesirable off-odours and microbial contaminants in dairy and bakery products
7. Detect undesirable, harmful, contaminants, toxics and taints
8. Search for chemical/biological weapons; for example, landmines buried in soil. Landmines elicit a tiny amount of explosive substances such as trinitrophenol. Taste sensors can easily be used to detect such explosives
9. Friend-foe identification
10. Leak detection

Taste recognition mechanism

Electronic tongues recognise taste using the same three levels as humans: the receptor level (taste buds in humans, lipid membrane of sensors in electronic tongues), the circuit level (neural transmission in humans, transducers in the electronic tongues) and the perceptual level (cognition in the thalamus in humans, statistical analysis by software in electronic tongues).

At the receptor level, the electronic tongue uses a sensor-probe assembly to detect organic and inorganic compounds. Each probe shows cross-sensitivity and selectivity, so each sensor could concurrently contribute to the detection of substances in the liquid matrix. These sensors have an organic coating that is sensitive to the species to analyse in the samples and a transducer that allows conversion of the membrane response into signals that will be analysed.

At the circuit level, the sample is quantified, digitised and results are recorded.

At the perceptual level, the statistical software interprets the sensor data into taste patterns. Depending upon the application, data analysis can produce a variety of information.

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